Monte Carlo Methods for Hypothesis Tests

Mark M. Fredrickson (mfredric@umich.edu)

Module 1: Integrate

Computational Methods in Statistics and Data Science (DATASCI 406)

Tests as Decisions

Estimators, Confidence Intervals, and Operating Characteristics

Last video: estimators $\hat{\theta}$ and confidence intervals $[A, B]$ are random variables, and everything we wanted to know about about them (**operating characteristics**) are **expectations over the sampling distribution**.

- Bias: $\mathrm{E}\left(\hat{\theta}\right)$
- Variance $\mathrm{Var}\left(\hat{\theta}\right)$
- Mean squared error: $\mathrm{E}\left((\hat{\theta}-\theta)^2\right)$
- Coverage: $P(A \leq \theta \leq B)$
- Expected width: $\mathrm{E}\left(B-A\right)$

Hypothesis testing.

A hypothesis test is a rule that turns a dataset into a **decision** between two options (H_0 and H_1). The **decision is a Bernoulli** random variable, and the two things we want to know about it are both probabilities:

$$\alpha = P(\text{reject} \mid H_0 \text{ true}), \qquad \text{power} = P(\text{reject} \mid H_1 \text{ true})$$

As ever, probabilities are expectations of indicators, so we can use MC to estimate these operating characteristics.

The pieces

- A **null hypothesis** H_0 and an **alternative** H_1, each naming a distribution for the data.
- A **test statistic** $T = T(X_1, \ldots, X_n)$: one number summarizing the evidence.
- A **rejection region**: the values of T that make us reject.

Two ways to be wrong:

	H_0 true	H_1 true
reject	**Type I error** (α)	correct
do not reject	correct	**Type II error** (β)

Size is α; **power** is $1 - \beta$. We fix the size we are willing to tolerate, then want as much power as we can get.

Why simulate at all?

For the classical tests you already know, c comes from a table: the null distribution of T has a name, so `qt` or `qnorm` hands you the cutoff.

That stops working the moment you leave the textbook cases:

- The statistic has no tractable null distribution (the maximum, a ratio, a distance between two curves).
- The distributional assumptions behind the table are false for your data.
- You don't want to use an asymptotic approximation.

Monte Carlo replaces the table. If you can *simulate* under H_0, you can build the null distribution yourself, and the cutoff follows.

The recipe

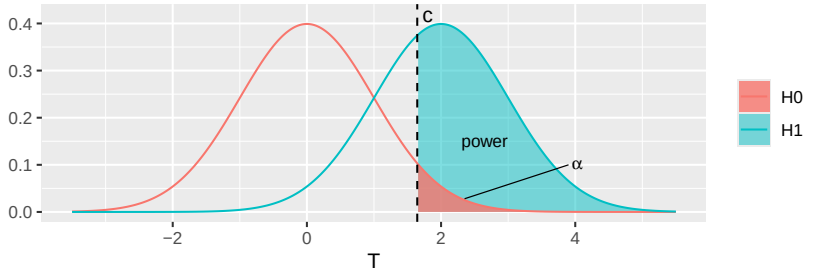
$$\alpha = \mathsf{E}\left(\mathbf{1}(T \in \mathcal{R}) \mid H_0\right), \qquad \text{power}(\theta_1) = \mathsf{E}\left(\mathbf{1}(T \in \mathcal{R}) \mid \theta_1\right)$$

Both are averages of indicators over simulated datasets:

```
null_dist <- replicate(k, T(rmodel(n, theta_0)))    # under H0
alt_dist  <- replicate(k, T(rmodel(n, theta_1)))    # under H1
tibble(H0 = null_dist, H1 = alt_dist) %>%           # look before you cut
  pivot_longer(everything()) %>%
  ggplot(aes(value, fill = name)) + geom_density(alpha = 0.5)
c_hat <- quantile(null_dist, 0.95)                  # the cutoff
mean(alt_dist > c_hat)                              # the power
```

Each of those is a sample mean or a sample quantile — except the picture in the middle, which is the one that tells you whether the rest are pointed the right way.

Plotting distributions



Two distributions of the **same statistic**, assuming two different hypotheses. Here we will use a simple cutoff for the rejection region, and the two shaded areas to its right are the two numbers we want:

- the sliver under the H_0 curve is the **size** α,
- the much larger area under the H_1 curve is the **power**.

Which values of T count as evidence against null?

Nothing forces $\mathcal{R}$ into the upper tail. Could be both tails or something else (tails are used when we have unimodal null) - H_1 shifts T **up**: $\mathcal{R} = \{T > c\}$, with c the upper α quantile of the null. - H_1 shifts T **down**: $\mathcal{R} = \{T < c\}$, with c the lower α quantile. A likelihood ratio L_0/L_1 runs this way. - H_1 can move T **either way**: split the budget, $\alpha/2$ into each tail.

There may be more than a single distribution for each hypothesis (**composite hypotheses**) requiring more plots.

Uniform with unknown maximum

Keep the setting from last video: $X_1, \ldots, X_n \overset{\text{iid}}{\sim} U(0, \theta)$, $n = 20$.

$$H_0 : \theta = 2 \quad \text{versus} \quad H_1 : \theta > 2$$

The two estimators from last video double as test statistics — both should be **large** when θ is large:

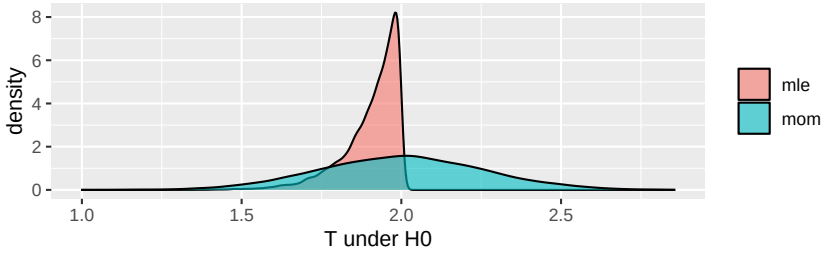
- $\hat{\theta} = \max_i X_i$ (the MLE)
- $\tilde{\theta} = 2\bar{X}$ (method of moments)

We know the MLE is the better *estimator*. That does not automatically make it the better *test*, and the two questions are not the same question.

Finding the Rejection Region

Simulating the null distribution

```
k <- 20000; n <- 20; theta0 <- 2
null_data <- replicate(k, runif(n, 0, theta0), simplify = FALSE)
null_mle <- map_dbl(null_data, max)
null_mom <- map_dbl(null_data, ~ 2 * mean(.x))
```



The cutoff is a quantile

We want the smallest region $T > c$ with $P(T > c \mid H_0) \leq 0.05$. That is the **upper 5% quantile** of the null distribution.

```
(c_mle <- unname(quantile(null_mle, 0.95)))
```

```
## [1] 1.995
```

```
(c_mom <- unname(quantile(null_mom, 0.95)))
```

```
## [1] 2.428
```

The MLE's cutoff is just **below** 2, and it has to be: $\max_i X_i < 2$ always under H_0, so a cutoff at or above 2 gives a test that never rejects anything, ever.

The method-of-moments cutoff sits well above 2, since $2\bar{X}$ is symmetric about 2.

Power

Power against a specific alternative

Simulate under H_1 and count rejections. Same code, different θ.

```
theta1 <- 2.2
alt <- replicate(k, runif(n, 0, theta1), simplify = FALSE)
c(mle = mean(map_dbl(alt, max) > c_mle),
  mom = mean(map_dbl(alt, ~ 2 * mean(.x)) > c_mom))
```

```
##    mle    mom
## 0.8560 0.2094
```

At the same size, the MLE test rejects a false null about four times as often.

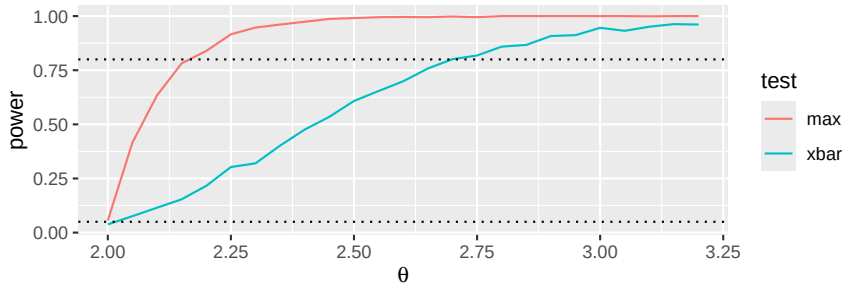
Power curves

One alternative is a single number. Usually we want power as a function of the alternative.

```
reject <- function(theta, k = 1000, n = 20) {
  X <- matrix(runif(n * k, 0, theta), nrow = n)
  c(max = mean(apply(X, 2, max) > c_mle),
    xbar = mean(2 * colMeans(X) > c_mom))
}
```

Both cutoffs are fixed: they came from the null simulation and do not move as θ does. Only the data generating X changes.

The two curves



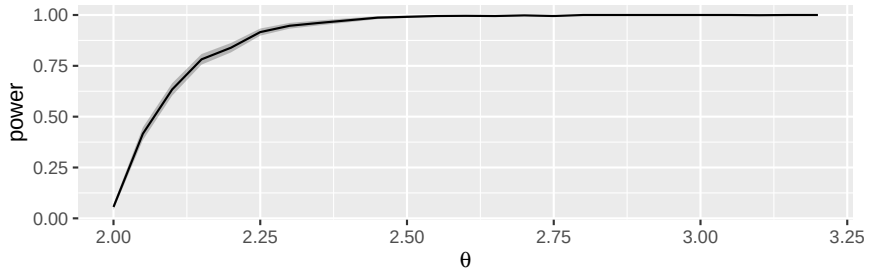
Reading the power curves

- **A power curve must start at α.** At $\theta = \theta_0$ the alternative *is* the null, so power is the size. A curve that starts above 0.05 is not a more powerful test; it is a broken one.
- Power rises to 1 as the alternative moves away. Every consistent test does this eventually; what separates them is **how fast**.
- `max` reaches 80% power while `xbar` is still near the floor.

Comparing tests means comparing **power at a common size**. A test with higher power and a larger size has not won anything.

Monte Carlo error on a power curve

Each point is a proportion out of $k = 1000$, so its standard error is at most $\sqrt{0.25/1000} \approx 0.016$.



Wiggles smaller than the band are noise. If a curve needs to be smooth, raise k — do not smooth the picture.

p-values

From a fixed cutoff to a measured one

A rejection region answers a yes/no question at a size fixed in advance. The ***p*-value** answers a finer one:

How extreme was the observed statistic, measured against the null distribution?

$$p = P(T \geq t_{\text{obs}} \mid H_0)$$

It is the smallest α at which this data would have led to rejection. And it is one more probability under H_0, so it is one more average of an indicator over the null simulation.

Computing a Monte Carlo p-value

```
x <- runif(n, 0, 2.4)     # one dataset from an alternative, "real data"
t_obs <- max(x)
mean(null_mle >= t_obs)
```

```
## [1] 0
```

Careful with the edge case. If `t_obs` exceeds every simulated null value this returns **exactly 0**, claiming the data is impossible under H_0. It is not impossible — we merely did not simulate enough.

Never report a p-value of zero

The standard fix counts the observed value as one of the draws:

$$\hat{p} = \frac{1 + \#\{T_b \geq t_{\text{obs}}\}}{B + 1}$$

```
mean(c(t_obs, null_mle) >= t_obs)
```

```
## [1] 5e-05
```

This can never be 0, it is never off by more than $1/(B+1)$, and it makes the test **exactly valid** at any size — the p-value is uniform under H_0, rather than merely approximately so.

The duality

A p-value and a rejection region are two views of one decision:

$$p \leq \alpha \iff T > c_\alpha$$

So you can work either way, and which you choose is a matter of what you want to report:

- The **region** is what you build when the size is fixed in advance and you run the test many times — a power study needs a yes/no on every replication.
- The p-**value** is what you report from a single analysis: it keeps information the yes/no throws away.

Two-Sided Tests

When the alternative can go either way

$$H_0 : \theta = 2 \quad \text{versus} \quad H_1 : \theta \neq 2$$

Now $\max_i X_i$ is evidence in **two** directions: too big says $\theta > 2$, too small says $\theta < 2$. So $\mathcal{R}$ has two pieces,

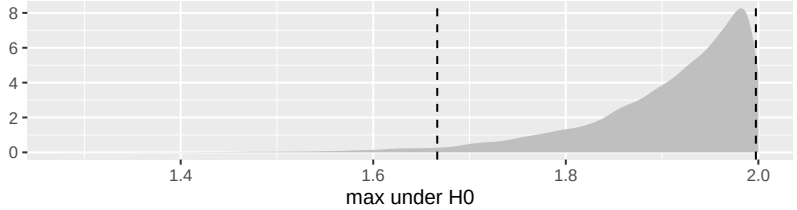
$$\mathcal{R} = \{T < c_{\text{lo}}\} \cup \{T > c_{\text{hi}}\}$$

and the 5% budget is split, 2.5% into each tail. Equal tails is a convention, not a law — but it is the right default when you have no reason to care more about one direction.

Two tails of very different sizes

```
(cs <- unname(quantile(null_mle, c(0.025, 0.975))))
```

```
## [1] 1.666 1.997
```



The lower piece is everything below 1.666; the upper piece is the sliver between 1.997 and 2. Each holds 2.5% of the null.

Two-sided does not mean symmetric. It means a fixed probability in each tail, and you get that by asking the null distribution for quantiles — never by taking $\pm$ something.

A two-sided p-value

Take the smaller tail and double it:

$$\hat{p} = 2 \min \{ P(T \leq t_{\text{obs}}),\ P(T \geq t_{\text{obs}}) \}$$

```
aug0 <- c(t_obs, null_mle)
p_lo <- mean(aug0 <= t_obs)
p_hi <- mean(aug0 >= t_obs)
(p_two <- min(1, 2 * min(p_lo, p_hi)))
```

```
## [1] 1e-04
```

Keep the $(1 + \#)/(B + 1)$ form in both tails, and cap at 1 — doubling a tail near 0.5 will otherwise hand you a p-value above 1.

Inverting a Test

The set of nulls you cannot reject

So far: fix θ_0, ask whether to reject. **Turn it around.** Hold the data fixed and ask which values of θ_0 survive the test:

$$\mathrm{CI}_{1-\alpha} = \{\theta_0 : \text{the level-}\alpha \text{ test of } H_0 : \theta = \theta_0 \text{ does not reject}\}$$

That set is a $1-\alpha$ confidence interval, and it needs no proof beyond the size of the test: if θ is the truth, the test rejects it with probability α, so the set contains it with probability $1-\alpha$.

Inverting on a grid

```
survives <- function(theta_c, t, B = 2000) {
  nd <- replicate(B, max(runif(n, 0, theta_c)))
  q <- quantile(nd, c(0.025, 0.975))
  t >= q[1] && t <= q[2]            # TRUE = do not reject
}
grid <- seq(2, 3.2, by = 0.01)
keep <- map_lgl(grid, survives, t = t_obs)
range(grid[keep])
```

```
## [1] 2.40 2.88
```

One simulated null distribution per candidate θ_0; the kept values are the interval.

Summary

- Size and power are **probabilities of rejecting**, so use Monte Carlo for expectations of indicators.
- Simulate null and alternative distributions. Find rejection with probability α under null with **high power** under alternative.
- May need plots. Fix null and sweep the alternative to get a **power curve**.
- Use the $(1 + \#)/(B + 1)$ form for Monte Carlo p-values: never zero, and exactly valid.
- **Two-sided**: 2.5% in each tail, taken as **quantiles** of the null. Not $\pm$ anything — the two pieces are the same probability, not the same width.
- **Invert** the test to get a confidence interval: the θ_0 you cannot reject. No pivot and no closed form required, so any test you can simulate gives you an interval.